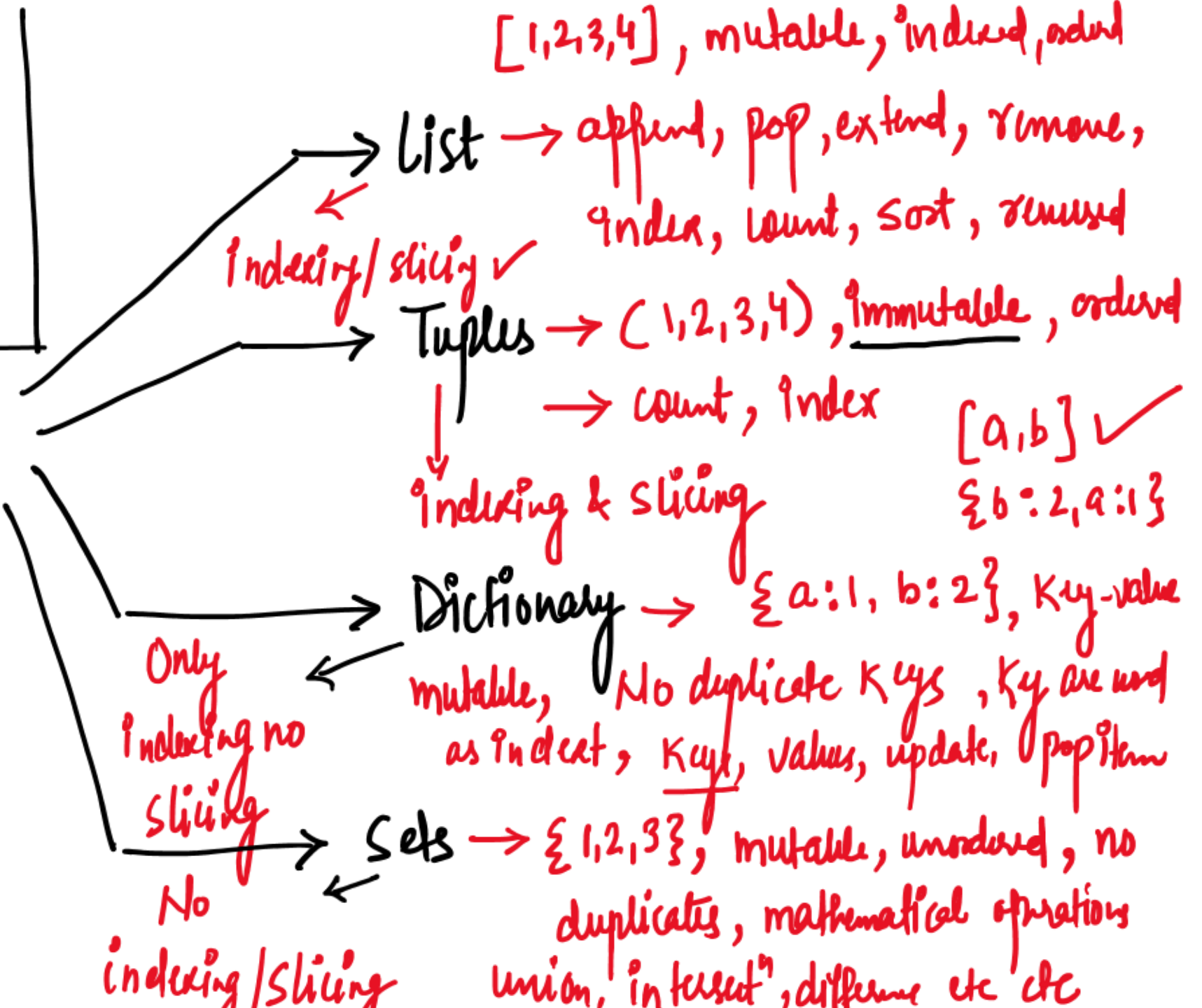


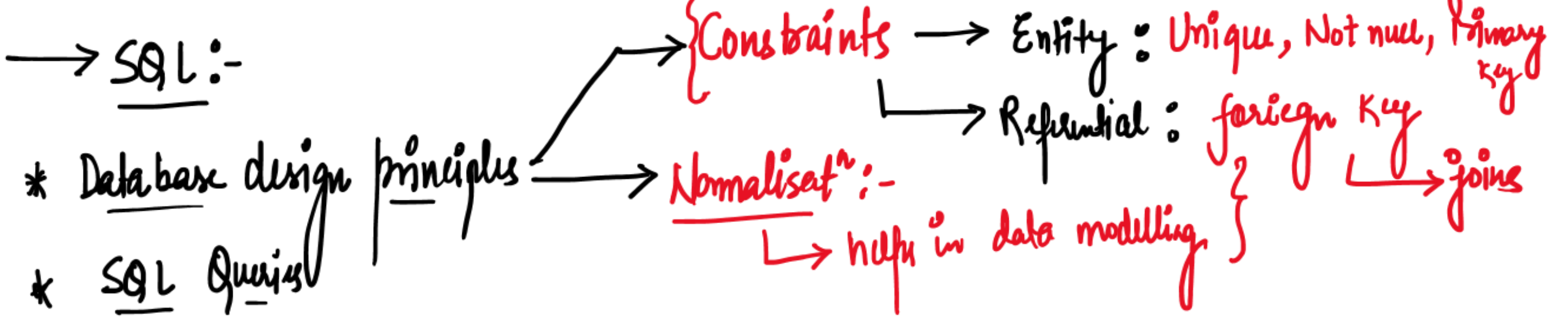
→ Python in-built functions:-

• print	• map	• range	• type
• len	• filter	• input	• list
• max	• zip	• round	• set
• min	• int	• del	• dict
• sum	• str/float	• sorted	• tuple

Python (Data Structures)

- String
 - Integer
 - float
 - bool
- Data types





→ * DOL - Create, Alter, drop, truncate

→ * DML - Insert, delete, update

→ * DAL - Select

→ * DCL - Grant, revoke

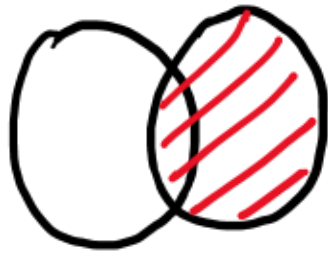
Truncate Table student ;

→ Joins:-

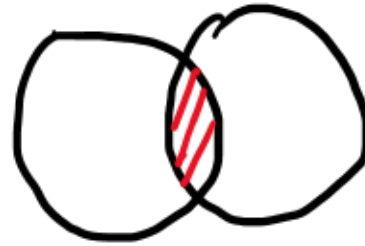
* ① Left Join



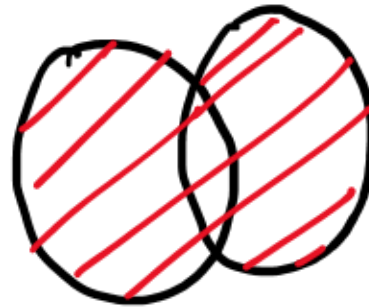
② Right Join



* ③ Inner join



④ Outer join / Union (SQL)



→ EDA (Exploratory Data Analysis) :-

Plotting graphs

Data + Problem Statement

Data visualisation

Analyse data

Decision making

→ Univariate : One variable

Histogram, distplot
pie chart

countplot, boxplot

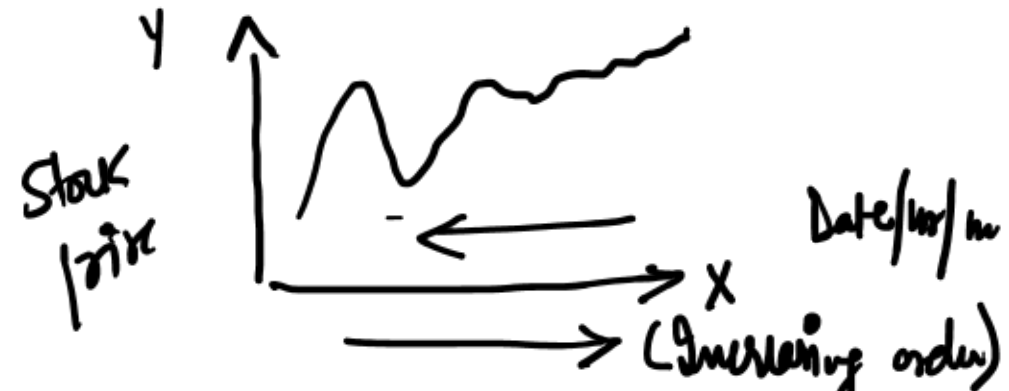
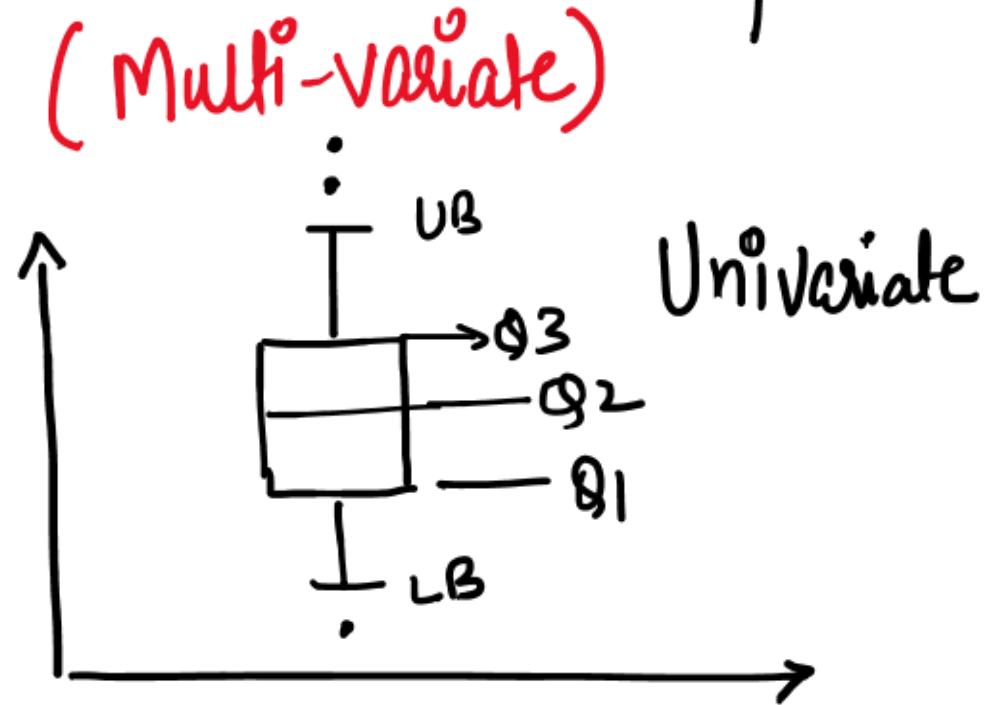
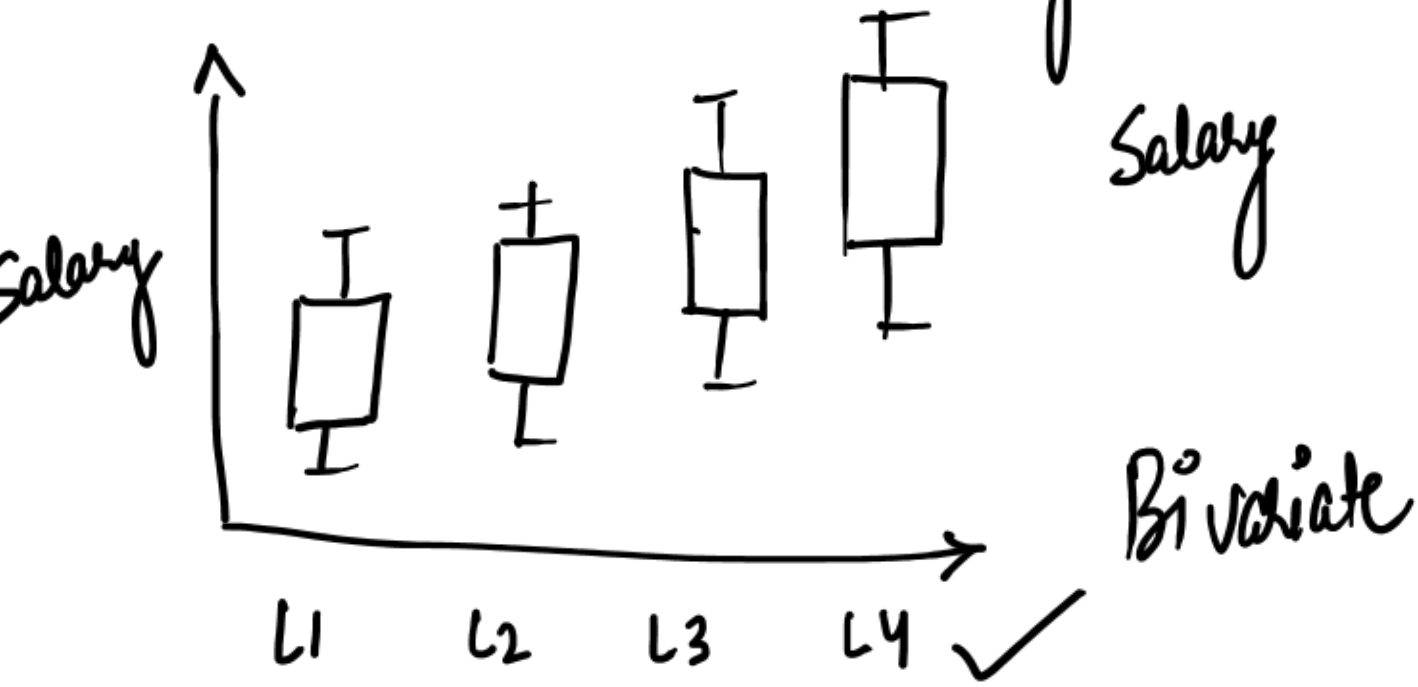
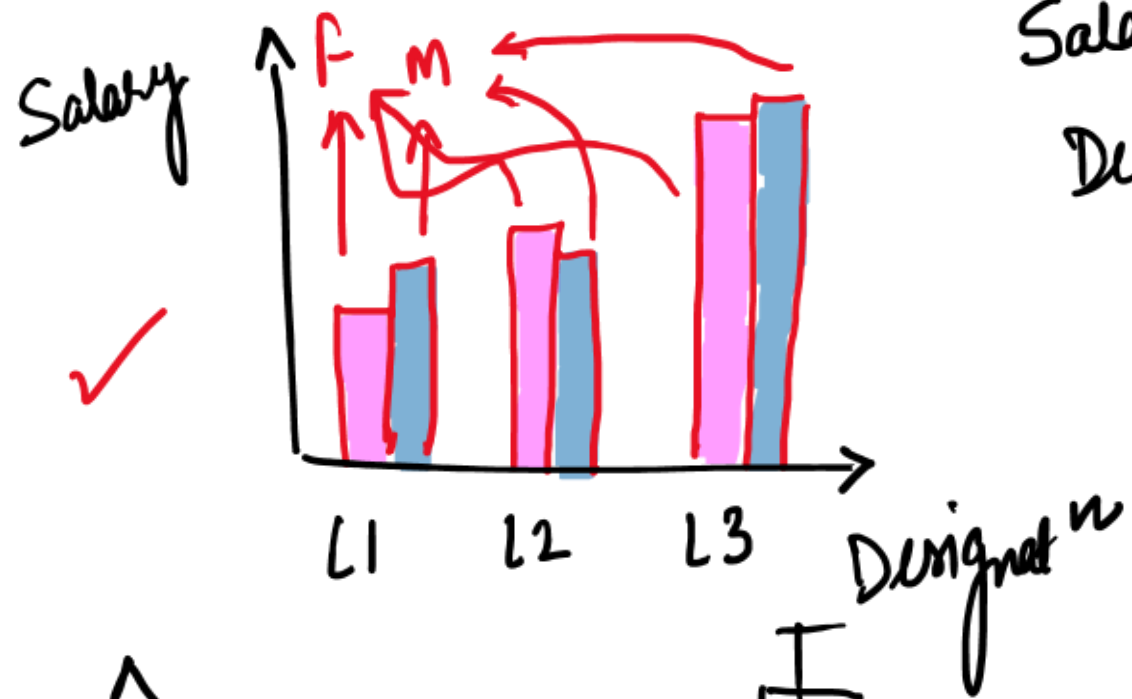
→ Bivariate : Two variables

Scatter, pairplot
Boxplot, barplot

→ Multivariate : Many variables

Heat maps

hue concept



→ Seaborn + Matplotlib

→ When to use | Syntax | how to read that graph

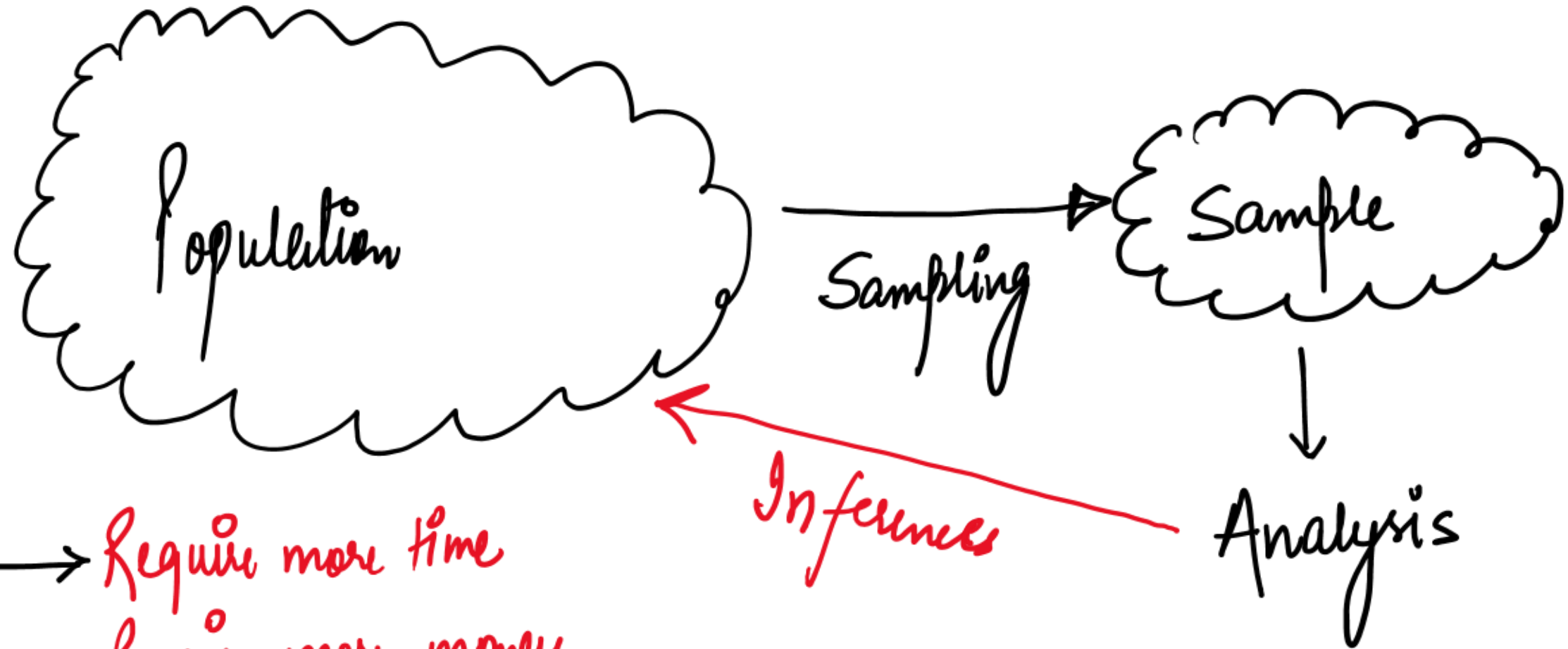
→ Missing values identificatⁿ + Treatment (if required)

→ Outlier identificatⁿ + Treatment (if required)

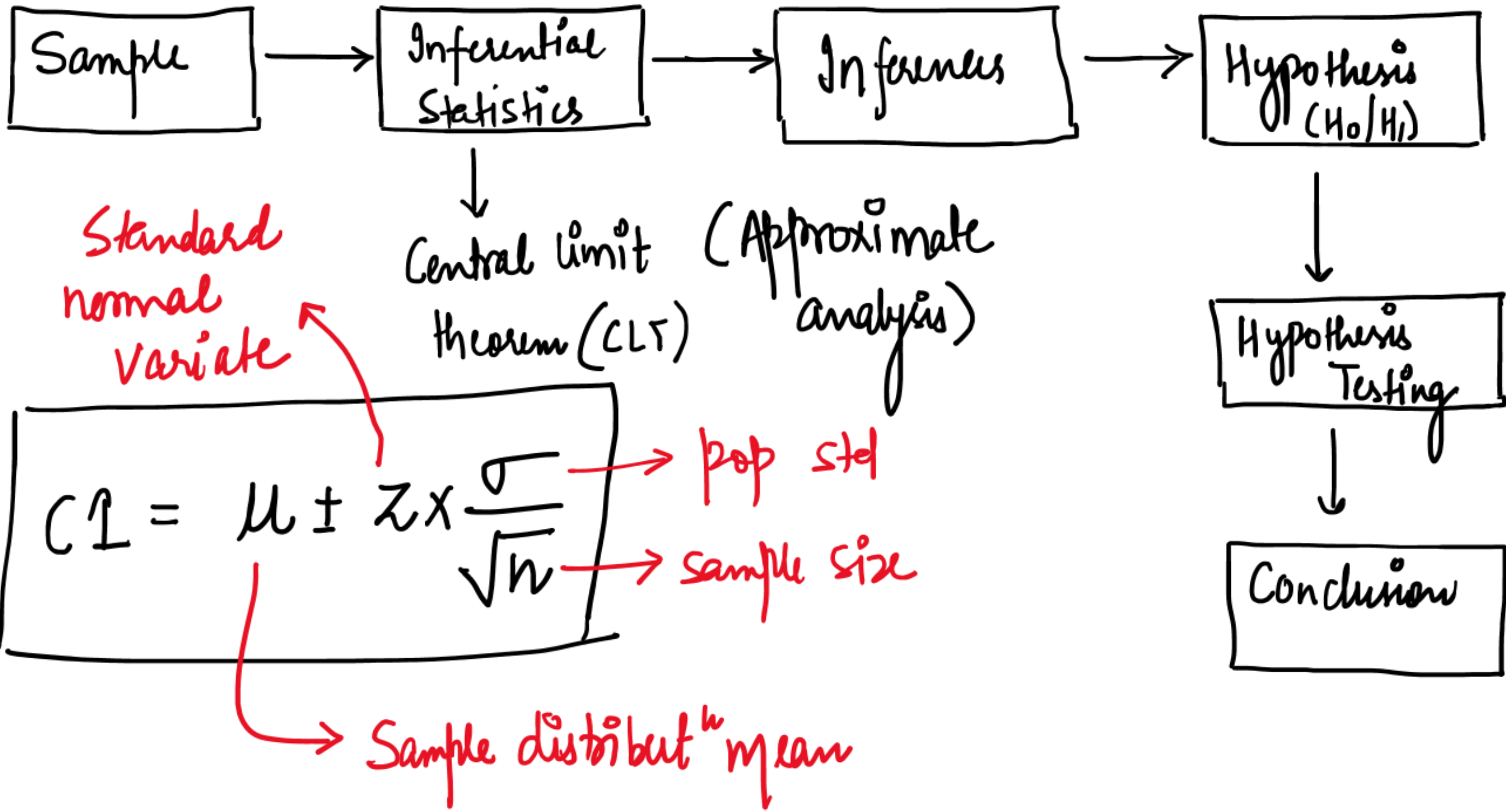
- Histogram : frequency plot (numerical columns)
- Distplot : Distributⁿ plot (N)
- Countplot : frequency plot (C)
- Piechart : % distributⁿ plot (C)
- Boxplot : Quantile Dist + Outliers (N | C-N)
- Line chart : Trend Analysis (N-N)
- Barplot : Relationship (C-C | C-N)
- Scatter plot : Relationship (N-N)
- Heatmap : Correlatⁿ Analysis (N-N)

- ① Load Dataset → `read_csv`, `read_excel`
- ② Explore & Understand data → `info()`, `describe`, `shape`, `columns`, `dtypes` etc etc
- ③ Data Cleaning → Remove duplicates | Removing absurd values | Correcting datatype
- ④ Segregate numerical & categorical columns → `select_dtypes`
- ⑤ (a) Univariate
 - `hist`
 - `dist`
 - `box`
 - Run loop on all numerical columns | `countplot` - categorical columns
- ⑥ Bivariate
 - N vs N → `Scatterplot`
 - C vs N → `Boxplot`
 - C vs C → `Boxplot`
- ⑦ Multivariate — `heatmaps` | `hue`

→ Statistical Inference :-



- Require more time
- Require more money
- Require more resources



→ $H_0: =, \geq, \leq$

→ $H_1: \neq, >, <$

Two-tail test: $=, \neq$

Left tail test: $\geq, <$

Right tail test:- $\leq, > \}$

Z-test $(n > 30)$

$\alpha = 5\%$ (Significance level)

SUPERMARKET LOYALTY PROGRAM

$$\mu = 120$$

$$n = 80$$

$$\bar{x} = 130$$

$$\sigma = 40$$

- A supermarket plans to launch a loyalty program if: Avg. spending per shopper > ₹120 per week
- A random sample of 80 shoppers enrolled in the pilot program spent an average of ₹130 in a week with a standard deviation of ₹40.

→ Should the supermarket launch the loyalty program?

$$(3) \text{ Z-stat} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{130 - 120}{\frac{40}{\sqrt{80}}} = 2.23$$

① formulation:-
 $H_0: \mu \leq ₹120$ X
→ $H_1: \mu > ₹120$ ✓

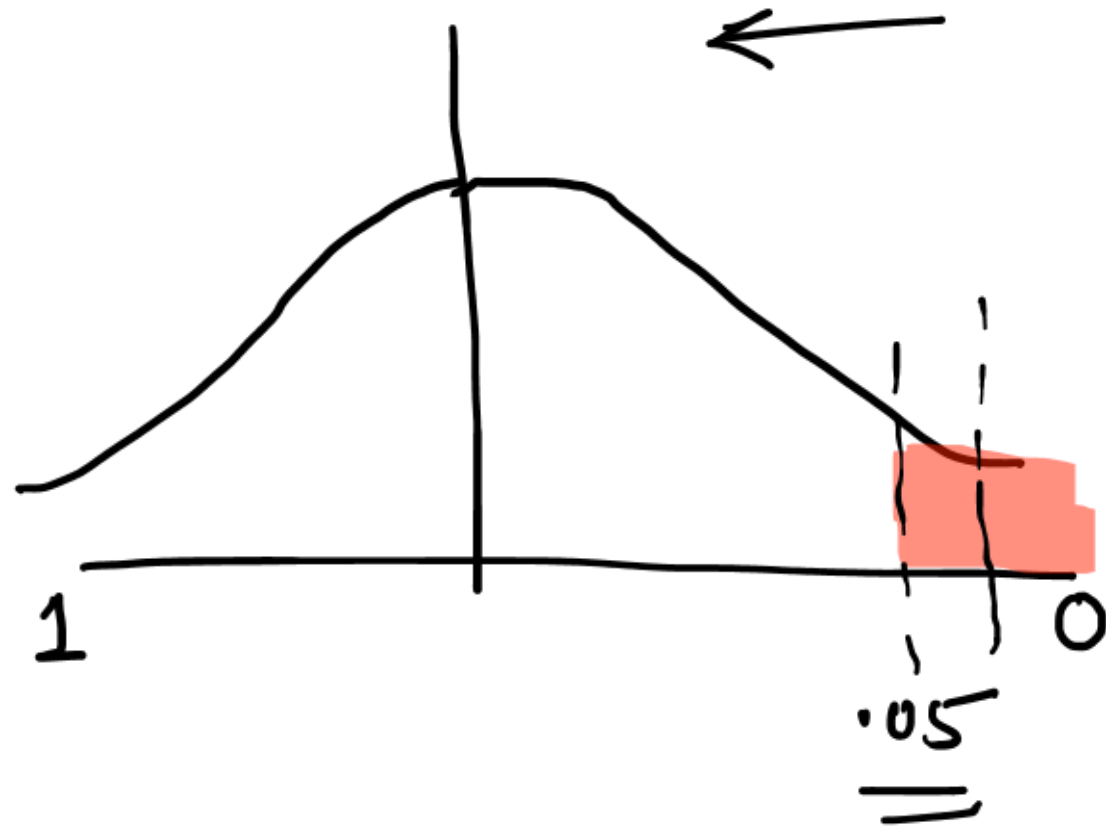
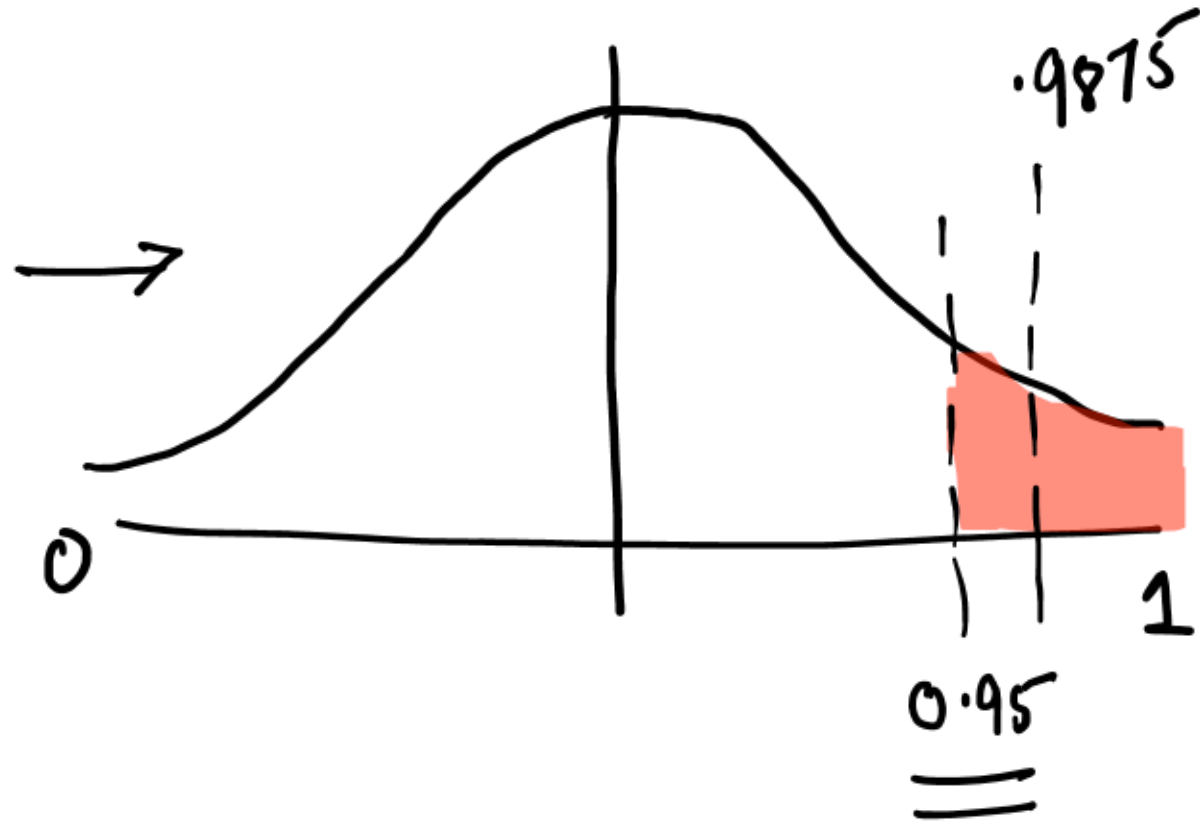
② which tail test:-

Right tail test

→ launch loyalty program

④ p-value: $\frac{0.9875}{1 - 0.9875} \rightarrow$ Reject null hypothesis

⑤ Decision:-



→ Anova Test :- (Analysis of variance) (mean of 3 or more groups different from each other)

→ Chi-Square test :- (Categorical variables)
↳ Association b/w two categorical columns

→ T-test :- t-distribution

* Paired — Same data in two scenarios

* Unpaired — Different data in different scenarios



{ SPAM FILTER

Spam % < 20%

- ABC company is considering buying a commercial filtering software costing \$15,000/annum that the vendor claims will reduce spam to less than 20%. ✓
- The company believes that if the software can reduce spam to less than 20%, then it can help improve employee productivity.
- The company collected a sample of 100 incoming mails with the spam filter and found that 11 of them were spam.
- Assume significance level $\alpha = 0.05$
- $\sigma = 0.4$

→ Should the company buy the spam filter?

Should buy

→ formulation :- $H_1: \mu < 0.2$ ✓
 $H_0: \mu \geq 0.2$ ✗ } left tail test

→ Z-stat: $\frac{x - \mu}{\sigma / \sqrt{n}} = -2.25$



→ p-value: 0.0122

Reject null hypothesis

→ Type I & Type II Error:-

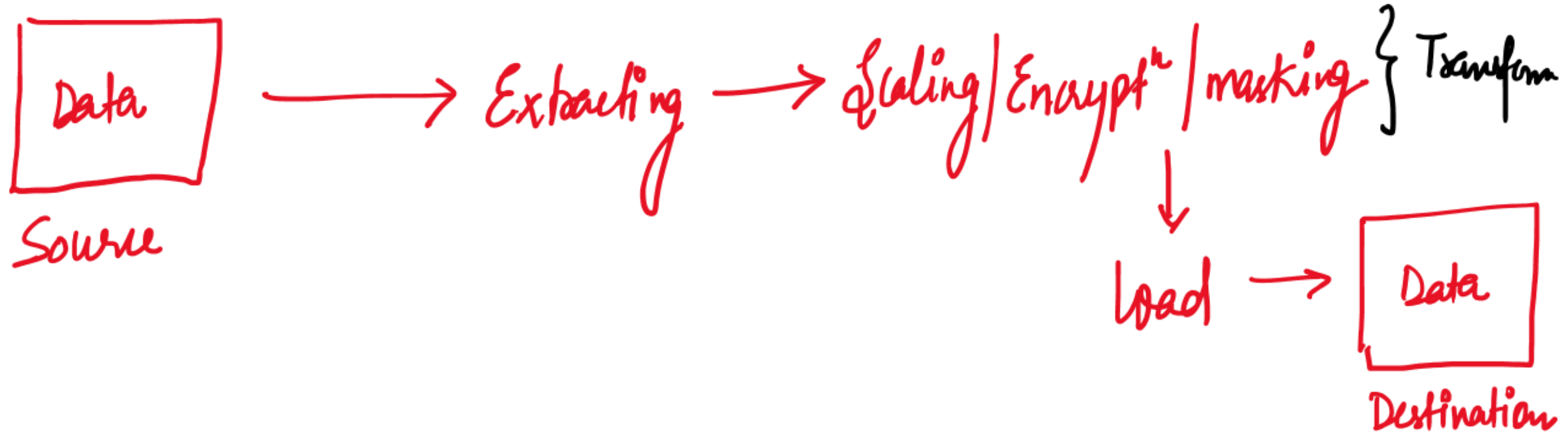


		True State of Nature	
		H_0 Is true	H_a Is true
Conclusion →	Support H_0 / Reject H_a	Correct Conclusion ✓	Type II Error
	Support H_a / <u>Reject H_0</u>	Type I Error	Correct Conclusion (Power)



→ ETL Operations:-

Extract — Transform — Load



→ Py-Spark :- Python API (library) for spark for
distributed data processing (Big data)

* In memory processing | Limits the use of disk

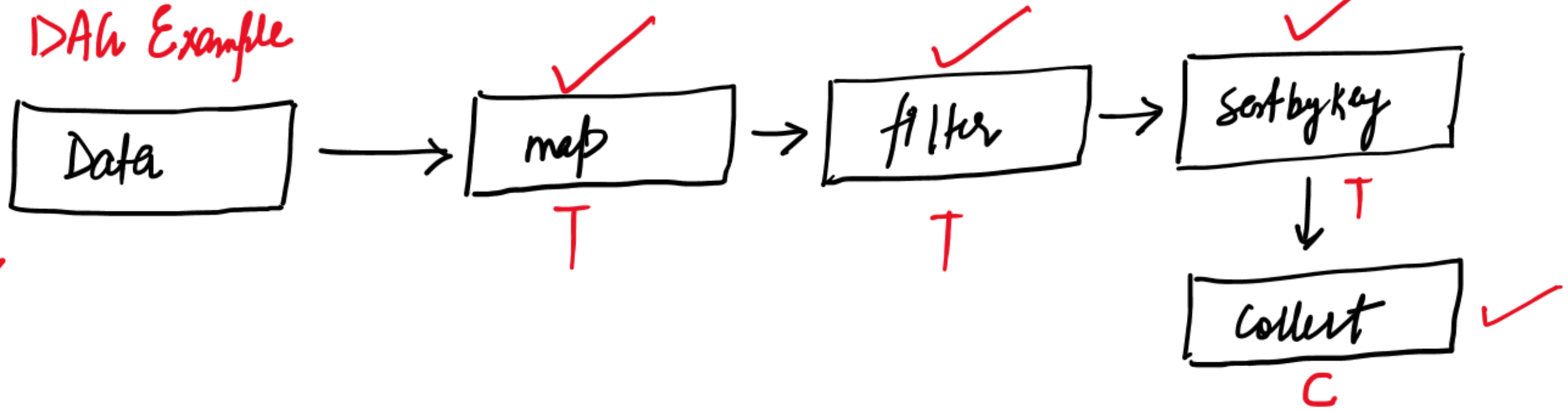
* fast & parallel execution

• Sql friendly }

→ lazy Evaluation :-

- * Transformations are not executed immediately.
- * Execution happens when an action is called.

DAh Example



→ Transformations (Lazy operations) [RDD, Dataframes]

⁺
paired RDDs (Key : value)

① Map : Applies f^n to each element

② filter : select elements

③ flat map : flatten output

④ Reduce by Key : Agg. using
Key

P.R ⑤ group by Key : group the values

P.R ⑥ sort by Key : sort the values

P.R ⑦ join : merge two RDDs

R ⑧ union : combine RDDs

→ Actions :- (Returns result & write the output)

① collect : Return data to driver

Executors → driver

② show : Prints the results

⑤ take(n) : to fetch n
number of elements

③ toPandas : to pyspark dataframe → pandas dataframe

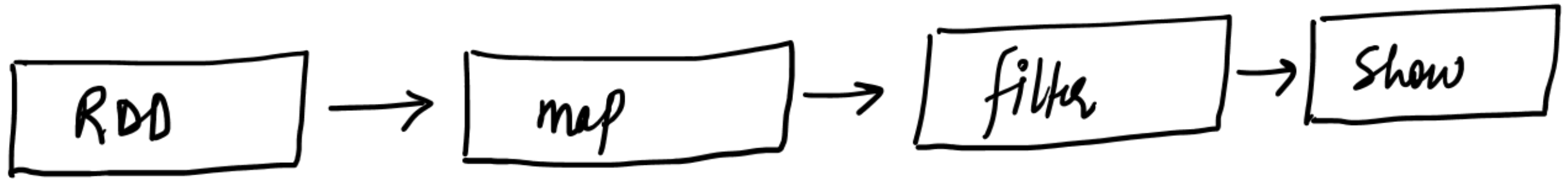
④ count : to count elements

→ Directed Acyclic Graph: (DAG)

Order / Sequence

No Loops

Executable Plan
from
code logic



(Brain)

Master Node

Primary node

Driver Program

Spark
Context

Resource
Allocation

Cluster
Manager

CPU, memory

Worker Node(s)

Task

Cache

Executors(s)

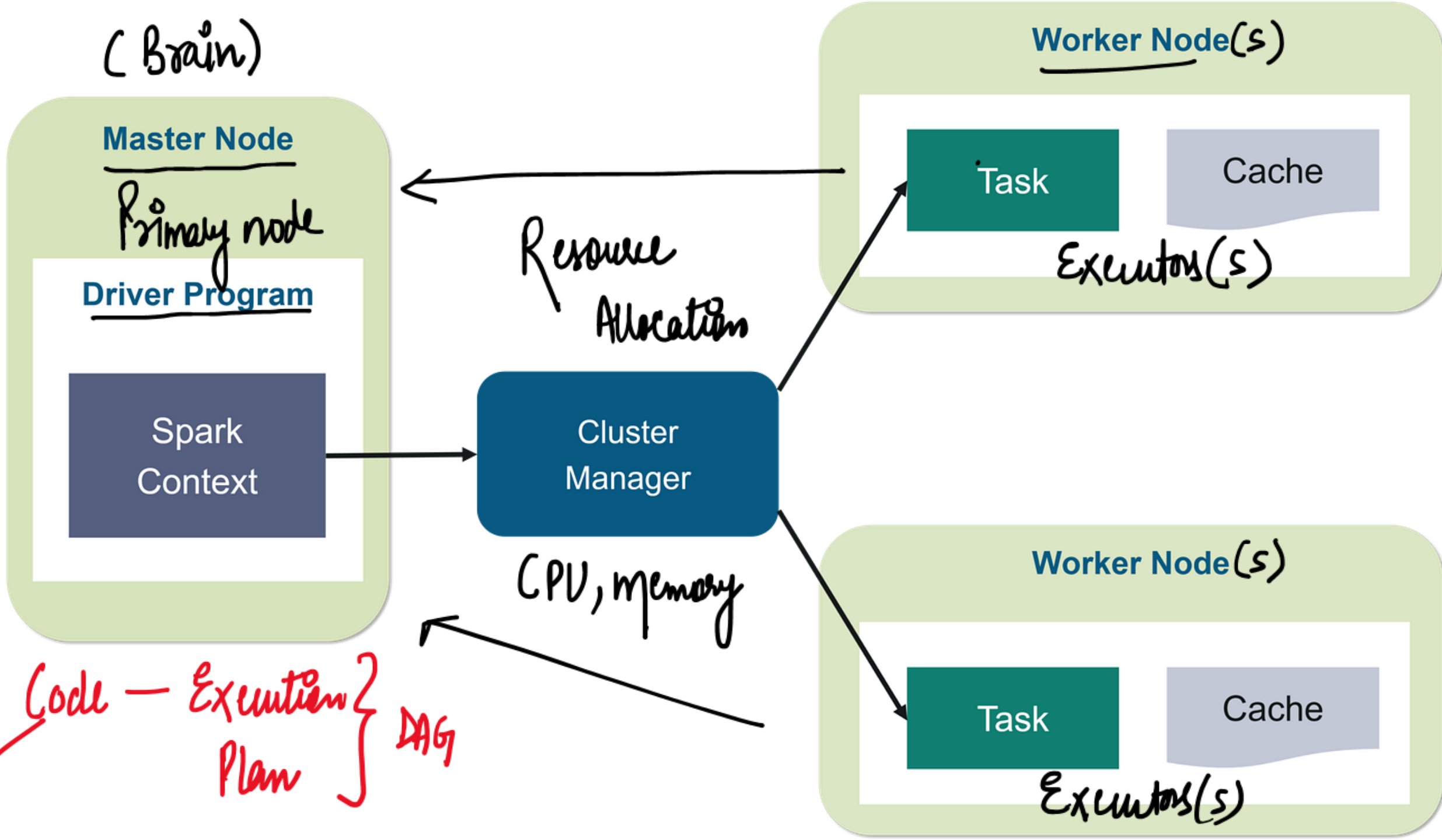
Worker Node(s)

Task

Cache

Executors(s)

✓ Code — Execution Plan } DAG



Cloud :- (virtual computer - servers) (AWS/GCP/Azure)

* Storage - S3, Google Cloud Storage

* Computing - EC2

* Database - RDS,

* Parallel processing - EMR, Databricks